FINAL REPORT

THE EVOLUTION OF THE ACADEMIC PERFORMANCE DURING THE YEARS

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SELECTED DATA

Introduction and Motivation

Education plays a central role in social and economic development, and educational indicators can help us understand how the academic profile of the population has changed over time. For this reason, our project focuses on the evolution of educational and labour-market indicators in Spain during the last decade. The main purpose is to analyse temporal trends, differences between Autonomous Communities and sex, and the possible effect of the COVID-19 period. To do this, we use different statistical techniques.

Description of the datasets

The data covers the period from 2013 to 2024 and refers to the 17 Autonomous Communities of Spain, with Ceuta and Melilla grouped together as an additional region. Therefore, the analysis includes 18 territorial units. The data were obtained from official education statistics and were organised into two CSV files: AC and F1. The AC dataset contains educational indicators, such as educational attainment and suitability rates, while the F1 dataset contains labour-market indicators, such as unemployment, employment and activity rates.

To build the AC dataset, we transformed the A1 Excel file, specifically section A1.2. We removed the rows corresponding to both sexes and kept male and female observations separately. We also removed total rows and added two variables, year and sex, to preserve this information. In addition, we included suitability rates for ages 12 and 15 from section C2.1, which measure the percentage of students studying the expected grade for their age.

To build the F1 dataset, we joined sections 1.4, 1.3 and 1.2 from the F1 Excel file. This dataset contains labour-market indicators: unemployment rates, employment rates and activity rates, grouped by education level and age group. As with the AC dataset, we removed total rows, kept male and female observations separately, and added year and sex variables. The year 2023 was omitted because the information was not available.

It is important to note that the data are aggregated at population level and do not contain individual student records. Therefore, there are no relevant privacy concerns. The data come from official publications, and the analysis is observational, since we are not conducting an experiment but using existing statistics from the Ministry of Education.

The AC dataset has 433 rows and 14 columns. Each row represents a combination of Autonomous Community, sex and year. The identification variables are Autonomous Community, sex and year, while the remaining variables are continuous percentages related to educational attainment and suitability rates. Educational attainment is divided into three levels: lower secondary, upper secondary and higher education, and each level is measured for different age groups: 25–64, 25–34 and 55–64. The dataset also includes suitability rates for ages 12 and 15.

The F1 dataset has 397 rows and 18 columns and follows the same observational structure as AC: each row represents an Autonomous Community, sex and year. However, instead of educational attainment, this dataset focuses on labour-market indicators. It includes unemployment rates, employment rates and activity rates. These variables are grouped by education level: lower education, middle education and higher education. Employment and activity rates are also divided into age groups such as 25–64 and 25–34.

Initial data analysis

During the initial data analysis, we found only four missing values in the F1 dataset. Since this represented a very small proportion of the data, we decided to remove them. Regarding variable types, Autonomous Community and sex were treated as categorical variables, while year was used as a temporal variable. The remaining variables were percentages or rates, so they were treated as continuous numeric variables.

LINEAR REGRESSION

After transforming and preparing the data, we built a linear regression model to predict the lower educational unemployment rate. We chose Lower_Unemployment_Rate as the response variable because the exploratory analysis showed relevant relationships between labour-market indicators and educational level. In particular, **Figure 1** shows a positive relationship between lower and higher education unemployment rates, suggesting that unemployment patterns are connected across education groups. This motivated us to study low-education unemployment in more detail using linear regression.

Using the cleaned dataset, we fitted an initial full model with all the available predictors. The main objective was to see how the variables behaved together and then reduce the model to make it simpler and easier to interpret.

Model selection

To select the best model, we first applied automatic backward selection using the step(direction = "backward") function. This procedure removes predictors step by step according to the AIC criterion. The result was a first reduced model, but it still contained many predictors, so we continued with manual backward elimination.

Using a significance level of $\alpha = 0.05$, we removed variables with high p-values one by one. After each removal, we compared the new model with the previous one using AIC and BIC. Lower values of AIC and BIC indicate a better balance between model fit and model complexity.

Finally, we checked multicollinearity using the VIF function. Predictors with very high VIF values were removed because they were strongly related to other variables. After this process, we obtained a simpler model with fewer predictors, while still keeping good explanatory power.

Model diagnostics and removing outliers

Before interpreting the final model, we checked whether the main assumptions of linear regression were reasonably satisfied. For this, we analysed the residual plots, the Q-Q plot, Cook's distance and the Residuals vs Leverage plot.

Some observations showed large studentized residuals or high Cook's distance, meaning that they could have a strong influence on the model. For this reason, we removed the most extreme observations and fitted the model again. After cleaning the data, we repeated the model selection process to make sure that the final model was not strongly affected by those influential points.

This step was important because the first diagnostic plots showed some deviations, especially in the upper tail of the Q-Q plot. Removing influential observations helped us obtain a more reliable model.

Hypothesis testing

We used hypothesis testing to analyse the statistical significance of the final linear regression model. First, we performed a global F-test. The null hypothesis stated that all regression coefficients were equal to zero, while the alternative hypothesis stated that at least one coefficient was different from zero.

The F-statistic was compared with the critical value of the F-distribution at $\alpha = 0.05$. Since the F-statistic was greater than the critical value, we rejected the null hypothesis. This means that the selected predictors explain a relevant part of the variability in Lower_Unemployment_Rate.

We also performed individual t-tests for the regression coefficients. These tests allowed us to analyse whether each predictor had a significant effect on the response variable. In the previous version, we mainly interpreted Year and Sex, but after the feedback we also paid attention to the rest of the relevant coefficients, looking at both their significance and the meaning of their estimates.

The variable Year showed that unemployment among people with lower education levels has generally improved over time. The variable Sex was also relevant, suggesting that gender differences play an important role in unemployment rates.

Prediction

To evaluate the predictive performance of the model, we divided the dataset into training and testing sets. Specifically, 70% of the observations were used to train the model, while the remaining 30% were used to test it.

The model was fitted using only the training data, and then we used the predict() function to estimate Lower_Unemployment_Rate for the testing observations. This allowed us to check how the model performed with data that had not been used during training.

In the updated version, we also included confidence intervals and prediction intervals. This helped us not only estimate the expected unemployment rate, but also understand the uncertainty associated with each prediction.

LOGISTIC REGRESSION

In this part, we wanted to work with a categorical response variable, so we decided to use a multinomial logistic regression model. Instead of predicting an exact numerical value, the objective was to predict the probability of each observation belonging to one of three employment performance categories: *LOW*, *MEDIUM*, or *HIGH*.

Before fitting the model, we transformed higher_employment_25_34 into a categorical variable called target_employment. To create this variable, we divided the observations into three groups using the 33rd and 66th percentiles. The lowest values were classified as LOW, the middle values as MEDIUM, and the highest values as HIGH.

Since the response variable has three categories, we used multinomial logistic regression instead of binary logistic regression. We selected LOW as the baseline category, so the model compares MEDIUM against LOW and HIGH against LOW.

Multinomial logistic regression and model selection
First, we fitted a full multinomial logistic regression model using all the available predictors. This model included educational variables, unemployment rates, employment rates, activity rates, sex, and year. The purpose of this first model was to have a starting point and see how all variables behaved together.

After fitting the model, we obtained the coefficients and standard errors. With them, we calculated Wald z-statistics and p-values to analyse which variables were useful for explaining the probability of belonging to each employment category.

As in the linear regression part, many variables were related to each other, so we also checked multicollinearity. Since vif() cannot be applied directly to a multinomial logistic regression model in the same way as in linear regression, we fitted auxiliary linear models with the same predictors. This allowed us to compute VIF values and detect variables with high collinearity.

Then, we started reducing the model step by step. First, we removed variables with high collinearity, comparing each new model with the previous one using AIC and BIC. After that, we continued the selection process using the p-values from the Wald tests.

For each variable, we checked the p-values in both comparisons: MEDIUM versus LOW, and HIGH versus LOW. If a variable had high p-values in both comparisons, we considered that it was not contributing enough to the model. We removed these variables one by one, always checking AIC and BIC after each change.

We stopped the process when the remaining variables had a p-value lower than 0.05 in at least one category, or when removing more variables could make the model worse. After this process, we selected the final reduced model, which was simpler than the full model and easier to interpret.

Hypothesis testing
To check if the reduced model was good enough, we performed a likelihood ratio test comparing the full model with the final reduced model.

The null hypothesis was: H0: The reduced model is sufficient.

The alternative hypothesis was: H1: The full model provides a significantly better fit.

The test statistic was computed using the difference between the deviance of the reduced model and the deviance of the full model. Then, we compared it with the critical value of the chi-squared distribution using a significance level of 0.05.

In our case, the p-value was greater than 0.05, so we did not reject the null hypothesis. This means that the reduced model is sufficient and that adding all the variables from the full model does not improve the model significantly. For this reason, we kept the reduced model.

Classification table and cut-off analysis

After choosing the final model, we evaluated how well it classified the observations. First, the model predicted the probability of belonging to each category: LOW, MEDIUM, and HIGH. Then, each observation was assigned to the category with the highest probability.

To evaluate the results, we created a confusion matrix comparing the real categories with the predicted ones. This allowed us to see how many observations were classified correctly and where the model made mistakes. We also calculated the overall accuracy, which represents the proportion of correctly classified observations.

Apart from accuracy, we computed sensitivity and specificity for each category using a one-vs-rest approach. This means that, for each class, we compared that class against the other two. Sensitivity measures how well the model detects observations that belong to a specific category, while specificity measures how well it detects observations that do not belong to it.

Finally, we plotted ROC curves for each category. These curves helped us compare sensitivity and specificity for different thresholds, and we selected the best threshold for each class.

Prediction of new observations

Finally, we used the final model to predict new hypothetical observations. These observations had different values of unemployment rates, employment rates, activity rates, sex, and year. The objective was to see how the model behaved with new cases.

For each observation, the model returned the probability of belonging to LOW, MEDIUM, and HIGH. Then, the predicted class was selected as the category with the highest probability.

The predictions included examples of the three possible categories, which showed that the model was able to distinguish between different employment performance situations. We also compared two observations with the same values except for sex. This allowed us to see how this variable affected the predicted probabilities, especially the probability of falling into the HIGH employment performance category.

Conclusions

In this third deliverable, we applied a multinomial logistic regression model to classify observations into three employment performance categories: LOW, MEDIUM, and HIGH. To do this, we transformed higher_employment_25_34 into a categorical response variable and fitted an initial full model with all predictors.

Then, we reduced the model step by step. First, we checked multicollinearity using auxiliary linear models and VIF. After that, we used Wald p-values, AIC, and BIC to remove variables that were not useful enough. This gave us a simpler and more interpretable final model.

The likelihood ratio test showed that the reduced model was sufficient, meaning that the full model did not provide a significantly better fit. Finally, we evaluated the classification performance using a confusion matrix, accuracy, sensitivity, specificity, ROC curves, and predictions for new observations.

Overall, this model allowed us to move from predicting a continuous unemployment rate to classifying employment performance levels. This gives us another perspective on the relationship between education and the labour market.

PCA

Overview
We performed a Principal Component Analysis on the merged dataset we created for the logistic regression model.

The goal is to reduce the dimensionality of a high-dimensional dataset and uncover the key underlying structures (primarily labor market and educational attainment) that explain the variance across observations. The analysis uses the **FactoMineR** and **factoextra** packages in R for computation and visualization.

Section 1: Data Preparation
In this section, we perform all the data preparation to create the data frame we will use for the PCA.

As the data we used for PCA is the same as the other parts, we selected the mixed data frame created in the linear regression section, mixing both datasets by CCAA, Sex and Year, omitting NA's and transforming sex into a factor. Then, renaming all variables for better understanding.

Section 2: PCA Setup
To compute PCA, we used the entire data frame, but the first three variables (autonomous_community, sex, year) were treated as supplementary using the quali.sup argument. This ensures they are excluded from the calculations of the PCA to create a pure numeric dataframe. However, we still give these variables as inputs to the function for further analysis to identify tendencies or clusters.

<u>Key arguments:</u>
quali.sup = c(1, 2, 3): Columns 1-3 (autonomous_community, sex, year) are declared as supplementary qualitative variables.
graph = FALSE: Suppresses automatic plot generation, plots are manually created later on the script.
scale.unit = TRUE: Standardizes all the variables to have the same standard so no variable with higher values dominates the analysis due to its magnitude.

Section 3: Dimensionality
Firstly we checked for the summary of the PCA. It prints eigenvalues, the percentage of variance explained by each component, and the cumulative variance. This is the primary tool for deciding how many dimensions to retain. Moreover, the summary also shows the contributions and the squared cosine for each dimension for variables and individuals, which we will analyse later on.

To determine the optimal number of dimensions, we analyzed the variance explained by each component. Although a clear 'elbow' was identified at the third dimension (setting the point where additional components stop contributing meaningfully) we opted to maintain only the first two. This decision is based on the fact that these two dimensions capture 71.9% of the total variance, providing a robust and interpretable 2D visualization of the underlying labor and educational structures.

The decision: Although 5 components have eigenvalues higher than 1, only the first two were kept for various reasons:

1. They explain 71.9% of the total variance.
2. After the third component, the successive dimensions contribute very little.
3. Two dimensions allow direct 2D visualization of the entire dataset.

Section 4: PCA Model Interpretation

Firstly, we extracted the most important values: eigenvalues (variance explained by each component), the coordinates of every individual for each dimension in the PCA space, and the variable correlations with each component. We projected the observations into the PCA space (**Figure 2**). By highlighting the observations by sex and adding confidence ellipses, we identified two clusters, confirming that sex is a primary driver of variance:

- **Female** observations cluster on the lower-left, also covering the upper left side (negative Dim 1 and negative/positive Dim 2).
- **Male** observations cluster on the upper-right (positive Dim 1 and positive Dim 2).

Afterwards, we also wanted to see if the passing of the years had some sort of effect. For that, we did the same as for sex but changing the variable to categorise to year (**Figure 3**). Here, another interesting output could be observed, unlike the clear-cut separation seen with sex, the year variable reveals a progressive transition across the PCA space. The ellipses trace a diagonal path from upper-left (earlier years, ~2013) to lower-right (later years, ~2024), suggesting a temporal trend towards improved employment and educational attainment over the study period.

Section 5: Quality and Contribution Analysis

The contribution of the variables can be seen in the two-dimensional space which can also be seen in a biplot (**Figure 4**). With this, we could more or less make an idea of what variables were contributing more to each component, looking at the direction of the arrow and the length. Moreover, this duality plot allows simultaneous interpretation of which variables drive group separation as the observations appear too. For example, employment rate arrows pointing towards the male cluster. Then, we also checked the percentages of the contributions of each variable to each dimension.

Afterwards, we checked the contribution of each variable to each dimension, to check if the dimensions were formed by a group of variables or not, we only focused on the first two dimensions.

For Dim 1, we found that it is dominated by employment and activity rate variables, particularly for the 25-64 age group. This dimension captures the overall level of labour market participation (we interpreted it as a Labour Market Dimension).

For Dim 2, we found that it is dominated by educational attainment and suitability variables (e.g., higher_education_25_34, age12_suitability, low_secondary_25_34). This dimension captures the population's level of educational qualification (we interpreted it as an Education Dimension).

Combining this to the clusters we found when plotting the individuals in the 2d space, some key findings could be interpreted:

- The male cluster shifts towards positive Dim 1, which reflects higher employment and activity rates.
- The female cluster shifts towards negative Dim 2 suggests higher rates of education level among females.
- The temporal pattern (years moving right and down) reflects improving employment (positive Dim 1) and rising educational attainment (negative Dim 2) from 2013 to 2024.

To validate the reliability of our interpretations, we analyzed the **squared cosine** (**Cos²**) values of the variables, which measure how well each observation or variable is represented by a component. A cos² close to 1 means the component captures almost all of the variance of that point or variable; close to 0 means it is poorly represented and should be interpreted cautiously.

The analysis showed that employment and activity rate variables (particularly for the 25-64 age group) are highly represented in Dim1 (but not that well educational variables), while educational and suitability variables are well-captured by Dim2.

Despite the general trend, an educational variable was not well captured by the "Education Dimension" we named before (Dim2), so we checked which dimension captured this variable better, which is Dim 3, and is poorly represented in the first two components.

Section 6: Outlier Detection

At the end, we did some outlier detection process, as in the biplot some points seemed to be a bit far away from the rest.

We made an outlier detection process focusing on observations with the largest Euclidean distances in the full PCA space (with all components). Larger distances indicate more atypical observations (those that deviate most from the average profile). Sorting in descending order surfaces the most extreme cases.

Observations with very high distances also appear visually as isolated points on the upper-left of the individual map, slightly separated from the main cloud.

The 5 most extreme cases were identified by index and then those indexes were used to identify the observations they belonged to, extracting the autonomous community, sex and year.

Finding: The most extreme outliers correspond to female observations from Ceuta and Melilla across multiple years (2013-2018). This autonomous city has a historically atypical socio-economic profile (characterised by very high unemployment and low activity rates) which could place these observations far from the average in PCA space. W hile their distances are large, they aren't the only ones far from the origin (other observations also exceed distances of 7-8), so no definitive outlier classification is made, they are considered as observations warranting separate contextual analysis.

CA

Variable selection

For the Correspondence Analysis(CA), we selected the variables *autonomous_community* and *low_unemployment_rate*. CA requires categorical variables, for that the unemployment rate variable was transformed into three categories using quantiles: Low, Medium and High. This method ensures that the observations are distributed more evenly between the categories. The resulting groups represent regions with low, medium and high unemployment levels. This analysis allows us to study if there are some Autonomous Communities associated with unemployment levels among people with lower educational attainment.

Contingency table

We create a contingency table that contains the frequency of observations associated with each combination of region and unemployment level. Also we use the chi-square test of independence to determine whether unemployment level and Autonomous Community are independent variables or not.

The obtained p-value was extremely small, therefore the null hypothesis of independence was rejected. This means that unemployment levels are associated with the Autonomous Community. The expected frequencies under the independence model were computed using the chi-square test. These values represent the frequencies that would be expected if unemployment level and Autonomous Community were completely independent.

By comparing observed and expected frequencies, we can identify which regions are overrepresented or underrepresented in each unemployment category.

Correspondence analysis

Correspondence Analysis was applied to project the relationship between Autonomous Communities and unemployment levels into a lower-dimensional space. This method makes it possible to visualize associations between regions and unemployment categories in a two-dimensional factor map.

The CA factor map shows the relative position of the Autonomous Communities and the unemployment categories. Categories located close to each other are interpreted as being associated. The first dimension mainly separates regions associated with high unemployment levels from those associated with low unemployment levels. See the **Figure 5**

Regions such as Ceuta y Melilla, Canarias, Andalucia, Extremadura and Castilla-La Mancha appear closer to the "High" unemployment category, indicating a stronger association with high unemployment levels among people with lower educational attainment. On the other hand, regions such as Aragón, Baleares, Rioja and País Vasco are located closer to the "Low" unemployment category, suggesting lower unemployment levels. Then for example Navarra and Asturias appear more associated with the "Medium" unemployment category.

Regions located near the centre of the graph, such as Catalunya or Cantabria, present a more average profile and are not strongly associated with a specific unemployment category.

Standardized residuals

The standardized residuals obtained from the chi-square test were used to support the graphical interpretation of the CA map. Positive residuals indicate that a region appears more frequently than expected in a given unemployment category.

The largest residuals were observed for Ceuta y Melilla, Canarias, Andalucia and Extremadura in the "High" unemployment category, confirming their strong association with high unemployment levels.

Conclusions

The Correspondence Analysis reveals that unemployment level and Autonomous Community are not independent. The results show the existence of important regional differences in unemployment among people with lower educational attainment.

Some regions are clearly associated with high unemployment levels, while others are closer to lower unemployment categories. Therefore, the analysis suggests the presence of strong territorial patterns in the Spanish labour market related to educational attainment.

CLUSTERING

In this section, we applied two clustering techniques — K-Means and Hierarchical Clustering — to two datasets: the original scaled dataset with all 26 variables, and the PCA-reduced dataset using the first two components, which together explain 71.9% of the total variance. This dual approach allows us to assess whether dimensionality reduction affects the quality and interpretability of the clusters.

Choosing the number of clusters

To select the optimal number of clusters, we applied both the elbow method (WSS) and the silhouette method to each dataset. The elbow method suggested K=4 as a reasonable choice for both datasets, where the rate of decrease in WSS begins to stabilize [**Figure 6**]. For the PCA dataset, the silhouette method clearly confirmed K=4 as the optimal value. For the original dataset, however, the silhouette method suggested K=2 [**Figure 7**], likely reflecting the male/female separation already identified in the PCA analysis. Since this division is already known and does not provide additional insight, we opted for K=4 in both cases to allow for a more meaningful and consistent comparison between the two approaches.

K-Means Clustering

Applying K-Means with K=4, the PCA dataset produced well-separated and spatially distinct clusters, while the original dataset showed considerable overlap along a diagonal band [**Figure8**]. This is consistent with the curse of dimensionality: with 26 correlated variables, K-Means struggles to find clean boundaries, whereas the PCA projection removes redundancy and exposes the underlying structure more clearly.

Examining the variable distributions per cluster, a clear socioeconomic gradient emerges across the four groups. Clusters differentiate primarily along education and employment dimensions: one cluster concentrates observations with low higher-education rates and high unemployment (a disadvantaged profile), while another shows the opposite pattern — high education attainment and high employment rates across all age groups. The mid-level variables show considerably less separation, indicating they are not the main drivers of the groupings. The age suitability variables (age12, age15) contribute to differentiating the remaining two clusters.

Hierarchical Clustering

We applied Ward's D2 linkage method to both datasets. The dendrograms confirm a natural partition into four groups, with clear height jumps at K=4 [**Figure 9**]. The scatter plots of the resulting clusters are visually very similar to those obtained with K-Means, which indicates that the four-cluster structure is robust and not an artifact of the algorithm chosen. [**Figure 10**]

Examining the variable distributions [**Figure 11**], the same socioeconomic gradient is present. However, compared to K-Means, the hierarchical approach produces tighter within-cluster distributions — the violin shapes are more compact — suggesting more internally homogeneous groups. Clusters 3 and 4 are more clearly differentiated here than in K-Means, particularly in the unemployment rate and age suitability variables.

Overall discussion

Both methods consistently show that the PCA space yields more interpretable and well-separated clusters than the original feature space. Across both techniques and both datasets, the four clusters reflect a spectrum from disadvantaged to high-performing autonomous communities in terms of education and employment, with unemployment rate acting as a secondary differentiator. The strong agreement between K-Means and Hierarchical Clustering reinforces the validity of the four-cluster structure found in the data.

FIGURES

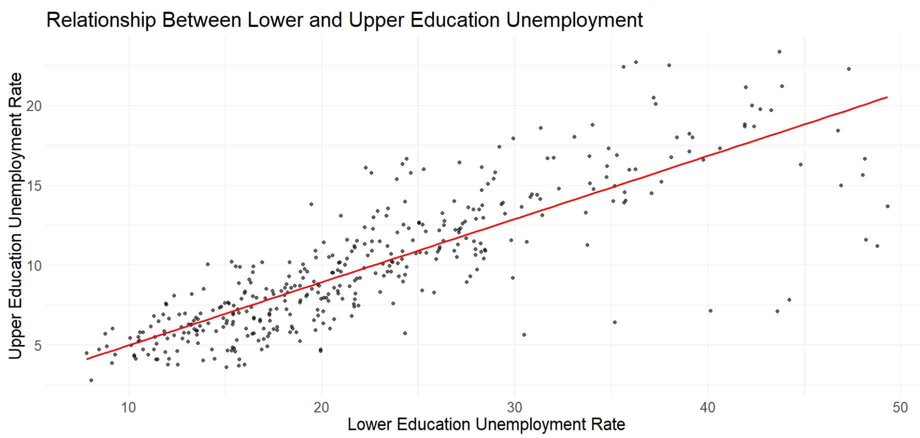

Figure 1

Figure 2

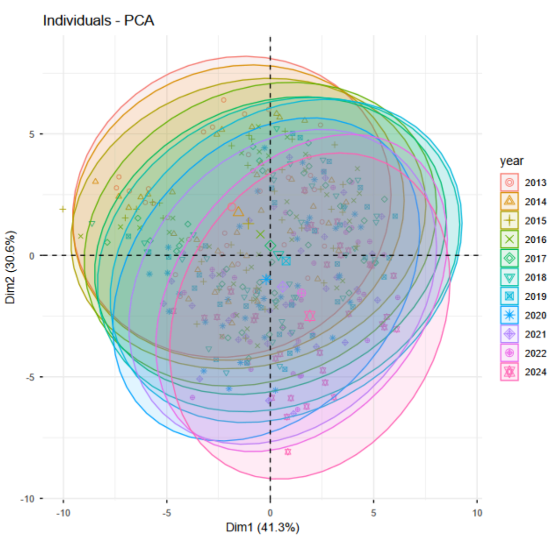

Figure 3

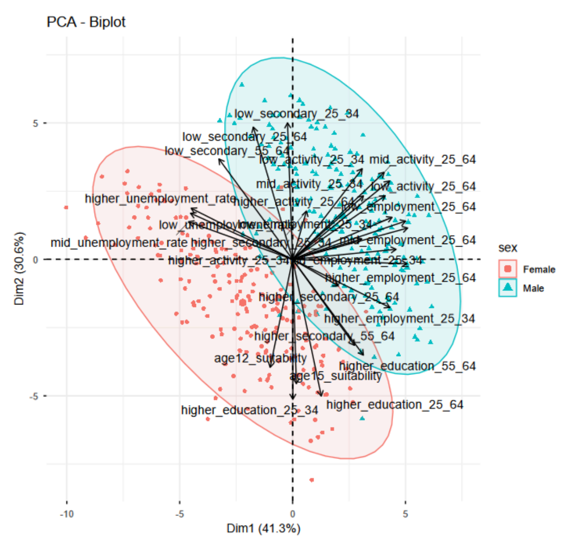

Figure 4

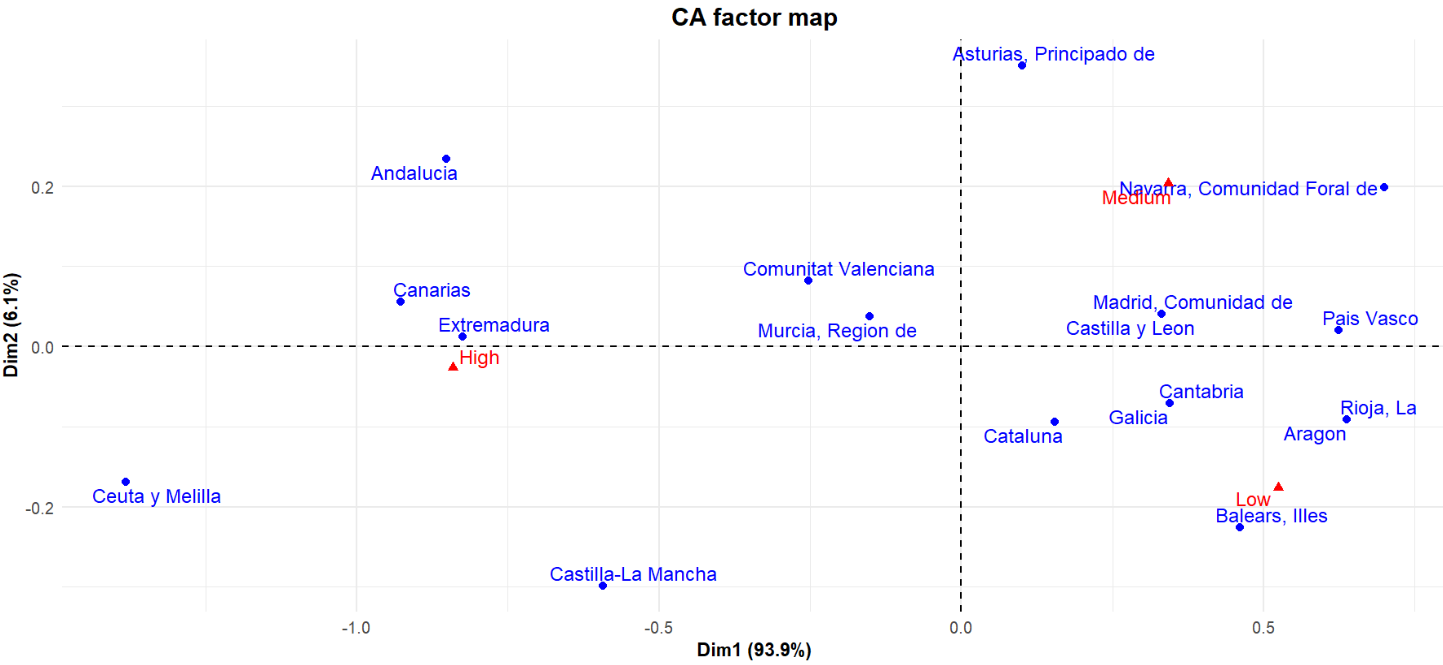

Figure 5

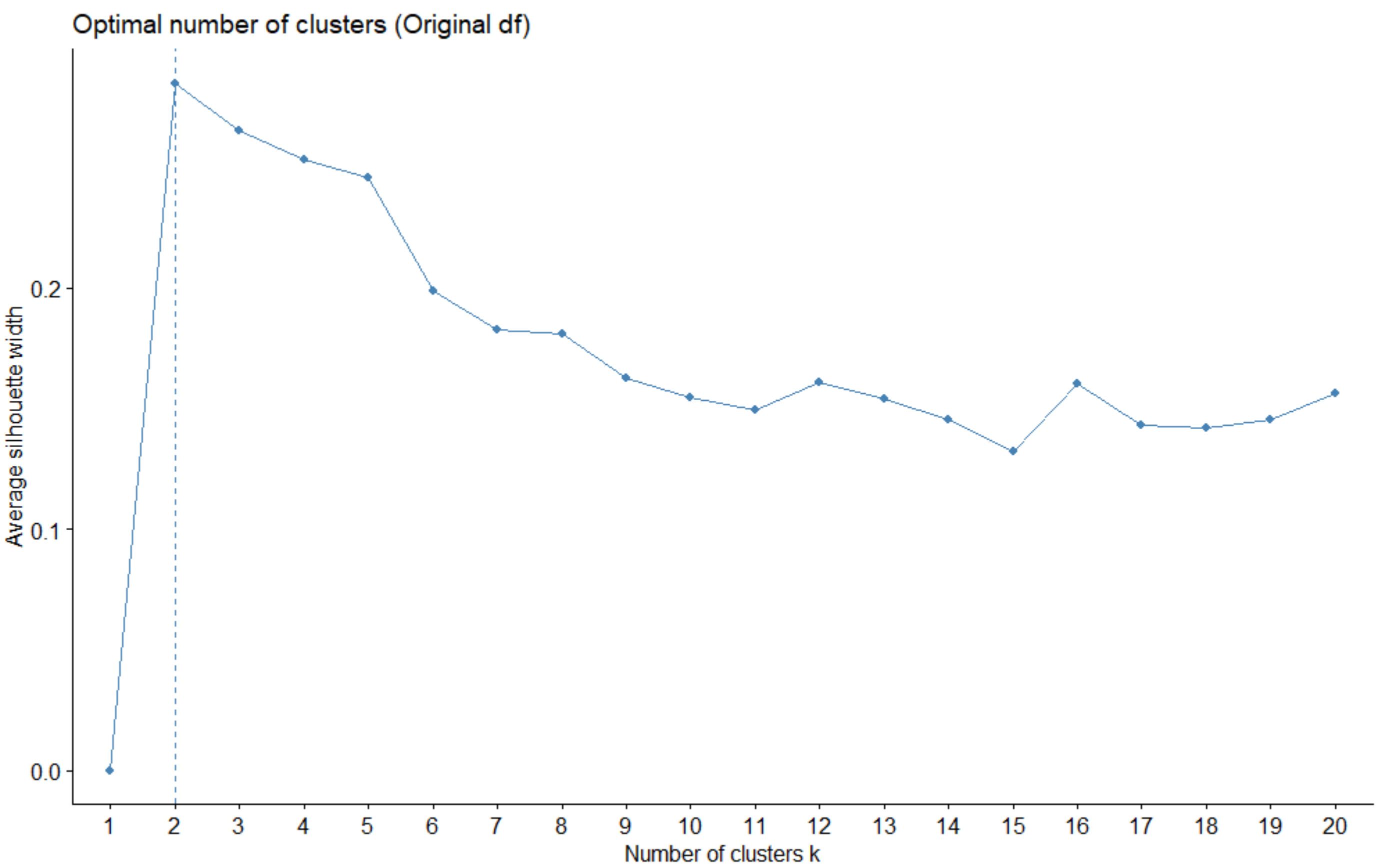

Figure 6

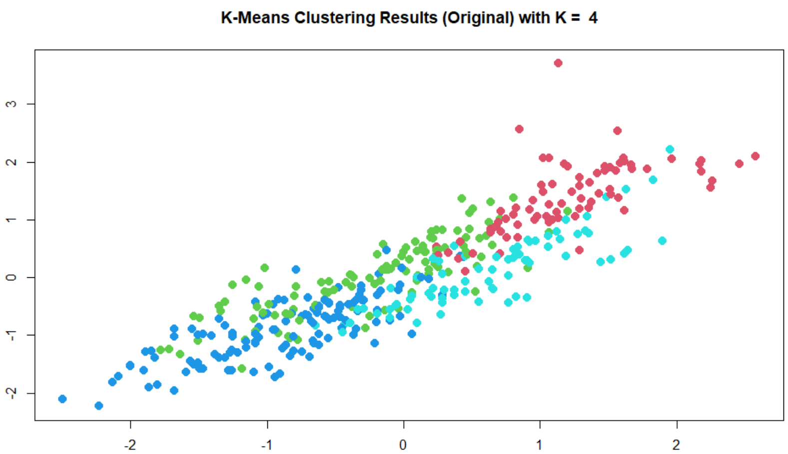

Figure 7

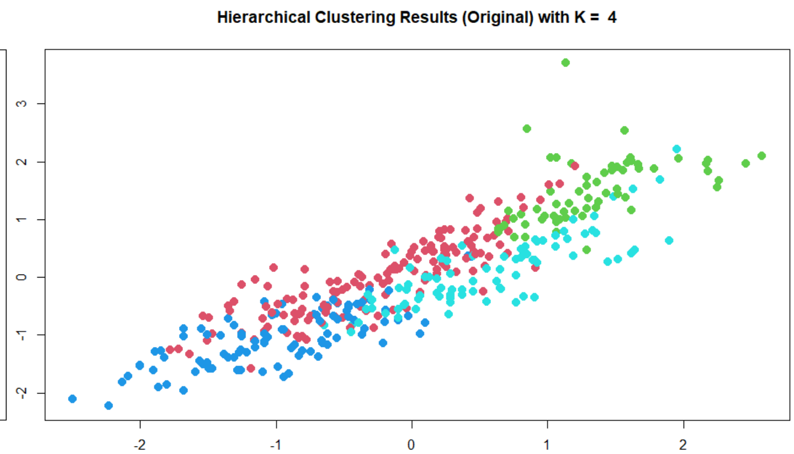

Figure 9

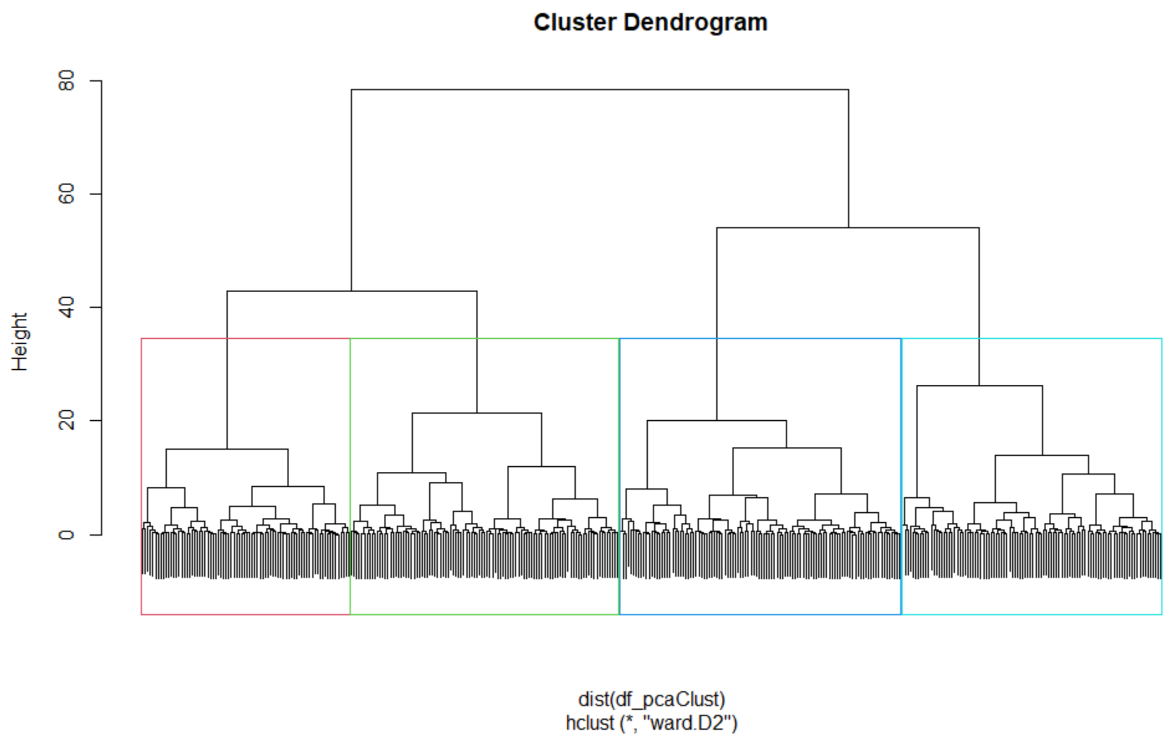

Figure 8

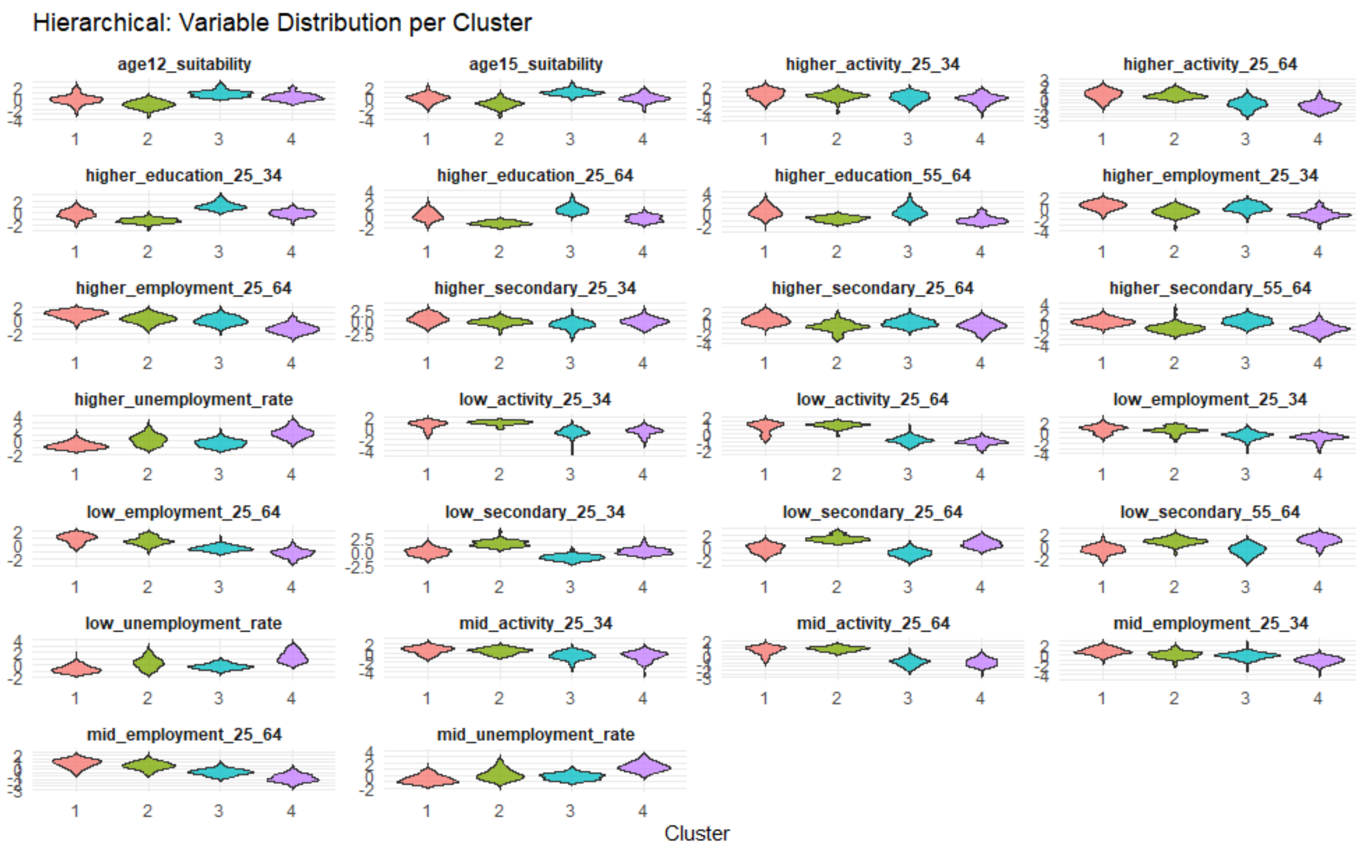

Figure 10

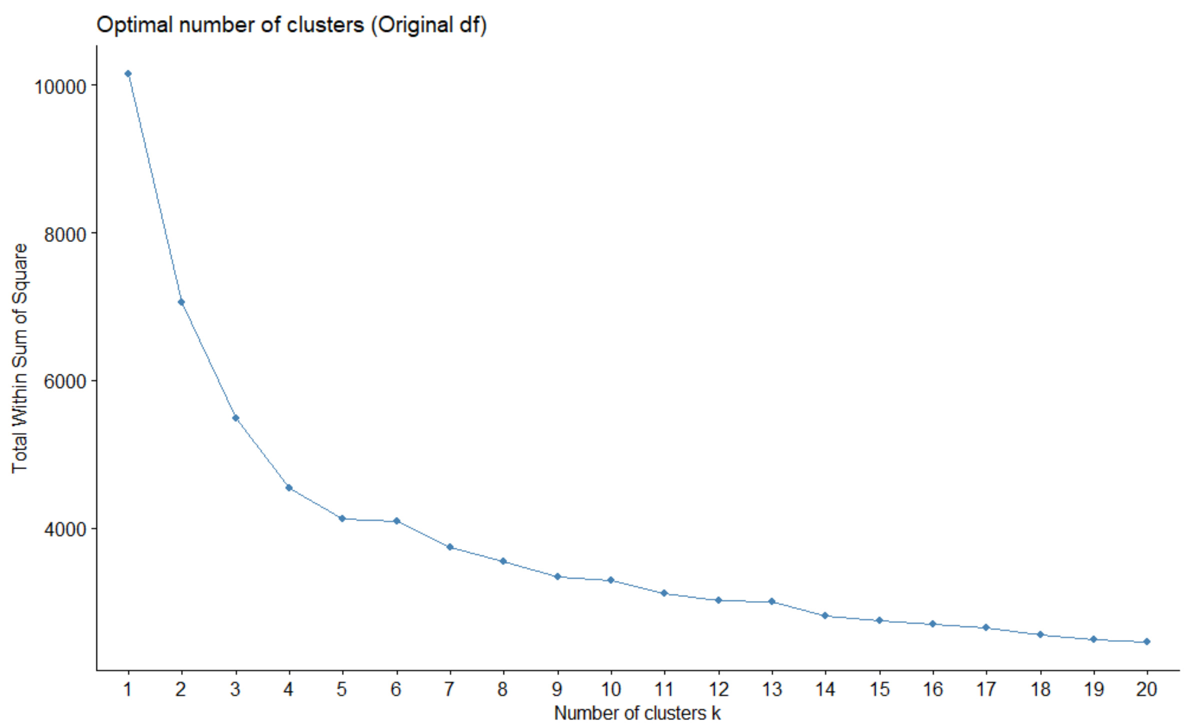

Figure 11